

NEEL GIRISH BAHADARPURKAR

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EDUCATION

M.S., Robotics Engineering - Worcester Polytechnic Institute (WPI)

Aug 2024 – May 2026

GPA 4.0 / 4.0 • Dr. Glenn Yee Graduate Student Tuition Award

B.Tech, Electronics & Telecommunications - K.J. Somaiya College of Engineering

June 2020 – May 2024

GPA 9.7 / 10

TECHNICAL SKILLS

Languages: C++ (Modern C++ 11/14/17), C#, Python, C, MATLAB, SQL

Planning, Robotics & Distributed Systems: motion & kinodynamic planning, sampling-based search (RRT, RRT*, EST, PRM), TAMP (PDDL), multi-agent coordination, real-time decision-making; OMPL, MoveIt, Nav2, ROS 1 / ROS 2; MQTT, Docker, JSON service contracts, distributed multi-process orchestration

Dev Tools, Testing & Simulation: Git, Linux, JIRA, Confluence; pandas, NumPy for statistical performance analysis; integration & benchmark harness design; Gazebo, MuJoCo, PyBullet, Genesis, Blender

RESEARCH EXPERIENCE

Graduate Research Assistant— Human-Inspired Robotics (HiRo) Lab, WPI

Jan 2025 – May 2026

Multi-agent planning and orchestration stack for a dual-arm mobile manipulator. C++, Python, ROS 2 / ROS 1, Jetson AGX Orin, MQTT, Docker.

- Architected an end-to-end ROS 2 planning + orchestration stack on Jetson AGX Orin coordinating perception, multi-agent decisions, and robot execution across distributed processes via containerized MQTT under sub-8s latency.
- Designed a four-LLM agent planning architecture (intent, dialog, task planner, verification) with a single-writer shared-state world model, enabling race-free coordination across decision layers and restart-tolerant deployment.
- Designed modular ROS service interfaces with JSON contracts bridging ROS 2 (Humble) and ROS 1 (Noetic) across networked machines; instrumented per-stage latency telemetry for bottleneck diagnosis.
- Built an automated 120-case integration test suite (99.2% planning success) before hardware deployment, then ran consecutive live demos on the IONA dual-arm platform; awarded the Dr. Glenn Yee Graduate Student Tuition Award.

PROJECTS

Motion & Kinodynamic Planning in C++ (OMPL)

Sep – Oct 2025

From-scratch Modern C++, OMPL, SE(2), statistical performance benchmarking.

- Implemented sampling-based and asymptotically optimal motion planners (RRT, RRT*, EST, custom) for a car with non-holonomic constraints from scratch in Modern C++ within OMPL with smart pointers, lambdas, STL, k-d trees, and nearest-neighbor queries across SE(2).
- Extended planners to kinodynamic regime with state-space constraints (velocity, acceleration, Dubins-car dynamics), validating trajectories against control-feasibility before execution.
- Designed an automated C++ benchmark harness running 80+ trials per configuration with statistical analysis of planning time, path length, and success rate; reusable collision-checking and solution-path modules with unit-test validation.

Task & Motion Planning for Robotic Manipulation

Oct – Dec 2025

PDDL, Pyperplan, OMPL, Genesis (Franka Panda 7-DOF), Python.

- Built a hybrid TAMP system integrating PDDL symbolic planning (Pyperplan) with OMPL motion planners via a symbolic↔geometric interface mapping abstract predicates to robot poses across planning layers.
- Designed an execution-replanning loop monitoring run-time state and re-grounding symbolic predicates from observations after each motion primitive, enabling closed-loop fault recovery for robust multi-step manipulation.
- Benchmarked four OMPL planners (RRT, RRT*, RRTConnect, PRM) on a Franka 7-DOF arm in Genesis with statistical analysis of planning-time vs. success-rate trade-offs to inform planner selection.

Einstein Vision - Multi-Model Perception Pipeline

Mar – Apr 2025

Mask R-CNN, YOLO, DETIC, Depth Anything V2, RAFT, Blender, PyTorch.

- Built a multi-model perception pipeline integrating Mask R-CNN, YOLO, DETIC, Depth Anything V2, and RAFT across segmentation, detection, open-vocabulary, depth, and optical flow with unified overlays.
- Generated synthetic scenarios in Blender with controlled lighting, geometry, and weather variations - a simulation-based validation pattern for stress-testing perception robustness across edge cases.
- Integrated vision foundation models (DETIC open-vocabulary detection, Depth Anything V2 monocular depth) into overlays, balancing per-frame latency against detection confidence to manage throughput/accuracy trade-offs.

Quadrotor Trajectory Tracking via Cascaded Control

Oct – Dec 2025

Crazyflie 2.0, cascaded PID, real-time control, Python.

- Implemented a real-time cascaded flight controller on a Crazyflie 2.0 drone with nested outer position, velocity and inner attitude, rate PID loops, fed by IMU + motion-capture state feedback for closed-loop 3D trajectory tracking on physical hardware.
- Achieved ~0.132 m 3D tracking RMSE (~0.04 m horizontal); applied per-axis statistical residual analysis to isolate altitude as the dominant error source, driving evidence-based gain-tuning trade-offs.
- Tuned loop rates and PID gains; built reproducible logging harness for statistical comparison across controller variants.